

Generated Certificates:

▼ certs2
 ca-key.pem
 ca.pem
 device-001-key.pem
 device-001.pem
 device-002-key.pem
 device-002.pem
 device-003-key.pem
 device-003.pem
 server-key.pem
 server.pem

Broker Startup:

PS C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA> cd "C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project"
PS C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project> mosquitto -c mosquitto_tls.conf -v
1773264355: Warning: The 'retry_interval' option is no longer available.
1773264355: mosquitto version 2.1.2 starting
1773264355: Config loaded from mosquitto_tls.conf.
1773264355: Bridge support available.
1773264355: Persistence support available.
1773264355: TLS support available.
1773264355: TLS-PSK support available.
1773264355: Websockets support available.
1773264355: Opening ipv6 listen socket on port 8883.
1773264355: Opening ipv4 listen socket on port 8883.

Broker receiving and sending data:

1773264383: No will message specified.
1773264383: Client HYDROLOGIC-Device-001 negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1773264383: Sending CONNACK to HYDROLOGIC-Device-001 (0, 0)
1773264384: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m1, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1773264384: Sending PUBACK to HYDROLOGIC-Device-001 (m1, rc0)
1773264387: New connection from ::1:54098 on port 8883.
1773264387: New client connected from ::1:54098 as GrandMarina-Dashboard (p4, c1, k60).
1773264387: No will message specified.
1773264387: Client GrandMarina-Dashboard negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1773264387: Sending CONNACK to GrandMarina-Dashboard (0, 0)
1773264387: Received SUBSCRIBE from GrandMarina-Dashboard
1773264387: hydroficient/grandmarina/# (QoS 1)
1773264387: GrandMarina-Dashboard 1 hydroficient/grandmarina/#
1773264387: Sending SUBACK to GrandMarina-Dashboard
1773264389: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m2, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1773264389: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m1, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1773264389: Sending PUBACK to HYDROLOGIC-Device-001 (m2, rc0)
1773264389: Received PUBACK from GrandMarina-Dashboard (Mid: 1, RC:0)
1773264394: Received PUBLISH from HYDROLOGIC-Device-001 (d0, q1, r0, m3, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1773264394: Sending PUBLISH to GrandMarina-Dashboard (d0, q1, r0, m2, 'hydroficient/grandmarina/device-001/sensors', ... (283 bytes))
1773264394: Sending PUBACK to HYDROLOGIC-Device-001 (m3, rc0)
1773264394: Received PUBACK from GrandMarina-Dashboard (Mid: 2, RC:0)
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Publisher with mTLS connection:

.....
HYDROLOGIC Sensor Publisher (mTLS)
=====
Device ID: 001
Topic: hydroficient/grandmarina/device-001/sensors
Certificate: C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\certs2\device-001.pem
=====
c:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\publisher_tls.py:108: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[TLS] mTLS configured with client certificate
[CONNECTING] localhost:8883...
[SUCCESS] Connected to broker as HYDROLOGIC-Device-001
[INFO] Certificate identity verified by broker
[PUBLISHING] Sending sensor readings (Ctrl+C to stop)...

Subscriber Recieving information as well as displaying:

```
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Grand Marina Security Dashboard (mTLS)
=====
Subscribing to: hydroficient/grandmarina/#
Certificate: C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\certs2\device-001.pem
=====
c:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\subscriber_tls.py:100: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
    client = mqtt.Client(client_id=CLIENT_NAME, **MQTT_CLIENT_ARGS)
[TLS] mTLS configured with client certificate

[CONNECTING] localhost:8883...

[LISTENING] Waiting for messages (Ctrl+C to stop)...

[SUCCESS] Connected to broker as GrandMarina-Dashboard
[INFO] Subscribing to: hydroficient/grandmarina/#
[SUBSCRIBED] QoS granted: (1,)

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
```

Key Test Rejection:

```
PS C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA> & C:/Users/gbemi/Downloads/python.exe "c:/Users/gbemi/OneDrive/Documents/ALL Projects/THE GRAND MARINA/Hydroficient Project/key test.py"
c:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA\Hydroficient Project\key test.py:11: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
    client = mqtt.Client(client_id="rogue-device", **MQTT_CLIENT_ARGS)
ERROR: Connection should have been rejected!
PS C:\Users\gbemi\OneDrive\Documents\ALL Projects\THE GRAND MARINA> █
```